

4th Estimation and Testing of hypothesis, large sample tests

Testing of hypothesis

1. small sample ($n < 30$)
2. large sample ($n > 30$)

large sample:

1. Single mean, We know that,

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \quad (\text{or}) \quad Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

2. Difference of mean,

We know that, $Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad (\text{or}) \quad Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

3. Single proportion

We know that, $Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$

Where,

p - sample proportion

P = population proportion

$$Q = 1 - P$$

4. Difference of Proportion

$$Z = \frac{P_1 - P_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$p = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

and $q = 1 - p$

Hint :

$\bar{x}$ $\rightarrow$ sample mean

μ $\rightarrow$ population mean

σ $\rightarrow$ standard Deviation

Sample : A finite subset of statistical individual in a population is called Sample.

Null Hypothesis : For apply the test of significance we first set up of a hypothesis a definite statement about the population parameter. Such a hypothesis is usually a hypothesis of no-difference and it is denoted by $H_0 : \mu = \mu_0$

Alternative hypothesis : Any hypothesis which is complementary to the null hypothesis is called alternative hypothesis, usually denoted by $H_1 : \mu \neq \mu_0$

Procedure for testing of hypothesis :

1. set up the null hypothesis
2. Choose the appropriate level of significance (either 5% or 1% level)

Level of significance (α) :

$\alpha = 5\%$ -1.96 to 1.96 } two-tailed
 $\alpha = 1\%$ -2.58 to 2.58 }

problems based on difference of mean

1. The means of two large samples of 1000 and 2000 numbers are 67.5 inches and ~~68.0~~ 68.0 inches respectively. Can the samples be regarded as drawn from the same population of standard deviation 2.5 inches.

sol Given that,

$$n_1 = 1000, n_2 = 2000$$

$$\bar{x}_1 = 67.5, \bar{x}_2 = 68.0$$

$$\sigma_1 = 2.5, \sigma_2 = 2.5$$

We know that, Difference mean

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$Z = \frac{67.5 - 68.0}{\sqrt{\frac{(2.5)^2}{1000} + \frac{(2.5)^2}{2000}}}$$

Same population of standard

1. $H_0: \bar{x}_1 = \bar{x}_2$ (No significant difference)

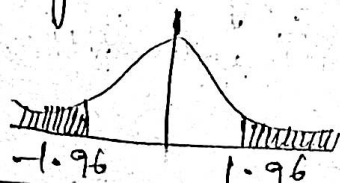
2. $H_1: \bar{x}_1 \neq \bar{x}_2$ (Significant difference)

3. $\alpha = 5\%$ (loss)

4. Test statistics

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{67.5 - 68.0}{\sqrt{\frac{(2.5)^2}{1000} + \frac{(2.5)^2}{2000}}} = 5.16$$

5. Critical region



6. Conclusion

$$|Z| = 5.16$$

5.16 (calculated value)

$$= 1.96 \text{ (table value)}$$

$$5.16 > 1.96$$

$\therefore H_0$ is rejected.

Note: Calculated value $>$ Table value

H_0 is accepted

Calculated value $>$ Table value

H_0 is rejected

2. A small sample of height 6400 English men has a mean of 1700 inches and standard deviation of 6.4 inches. While a sample of heights of 1600 Australians has a mean of 172 inches and a standard deviation of 6.3 inches. To the data indicate the Australians are on the average taller than English men.

So Given, $n_1 = 6400$; $n_2 = 1600$

$$\bar{x}_1 = 1700; \bar{x}_2 = 172$$

$$\sigma_1 = 6.4 \quad \sigma_2 = 6.3$$

We know that,

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

1. $H_0 = \bar{x}_1 = \bar{x}_2$ (No significant value)

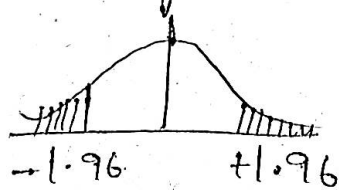
2. $H_1 = \bar{x}_1 \neq \bar{x}_2$ (significant different)

3. $\alpha = 5\%$ (los)

4. Test statistics

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{170 - 172}{\sqrt{\frac{(6.4)^2}{6100} + \frac{(6.3)^2}{1600}}} = -11.32$$

5. Critical region



6. Conclusion

$$|Z| = 11.32$$

± 1.32 (calculate value)

± 1.96 (table value)

$$11.32 > 1.96$$

$\therefore H_0$ is rejected

Problems based on Single mean

1. A sample of 900 members has a mean of 3.4 cms and standard deviation 2.61 cms. If the sample from a large population of mean 3.25 cms and standard deviation 2.61 cms (Test at 5% los)

Q1 Given, $n = 900$ members

$$\bar{x} = 3.4$$

$$\sigma = 2.61$$

$$\mu = 3.25$$

We know that,

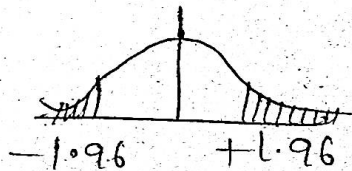
$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

1. $H_0: \bar{x}_1 = \mu$ (No significant difference)
2. $H_1: \bar{x} \neq \mu$ (Significant difference)
3. $\alpha = 5\%$ (los)

4. Test statistics

$$\begin{aligned}
 z &= \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{3.4 - 3.25}{\frac{2.61}{\sqrt{900}}} = \frac{3.4 - 3.25}{\frac{2.61}{30}} = 3.4 \\
 &= \frac{3.4 - 3.25}{0.087} = \frac{0.15}{0.087} = \underline{\underline{1.724}}
 \end{aligned}$$

5. Critical region



6. Conclusion

$$|z| = 1.724$$

$$= 1.96 \text{ (table value)}$$

$$1.724 < 1.96$$

$\therefore H_0$ is accepted

2. A Sample of 100 students is taken from a large population a mean height of the students in sample is 160 cm can it be reasonably regarded in the population the mean height is 165 cm and a standard deviation is 10 cm at 1% level of significance

sol

$$n = 100 \quad \mu = 165$$

$$\bar{x} = 160 \quad \sigma = 10$$

5

We know that,

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

1. $H_0 : \bar{x} = \mu$

2. $H_1 : \bar{x} \neq \mu$

3. $\alpha = 1\%$

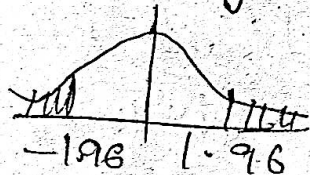
4. Test statistics

$$\bar{Z} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{160 - 165}{\frac{10}{\sqrt{100}}}$$

$$\bar{Z} = \frac{160 - 165}{1} = -5$$

$$\bar{Z} = -5$$

5. critical region



6. Conclusion

$$|Z| = |-5|$$

$$|Z| = 5$$

$$= 1.96 \text{ (table value)}$$

$$5 > 1.96$$

$\therefore H_0$ is ~~accepted~~ rejected

problems based on Difference of proportion

1. Before an increase in exercise duty on tea 800 persons out of a sample of 1000 persons were found to be tea drinkers. After an increase duty 800 people were tea drinkers in a sample of 1200 people. Using standard error of proportion state whether there is a significant decrease in the consumption of tea after the increase in exercise duty (z_{α} at 5% l.o.s. 1.645 / 1% level 2.33)

sol Given that,

$$n_1 = 1000 \quad n_2 = 1200$$

$$p_1 = \frac{800}{1000} \quad p_2 = \frac{800}{1200}$$

1. $H_0 : P_1 = P_2$ (No significant difference)
2. $H_1 : P_1 \neq P_2$ (Significant difference)
3. $\alpha : 5\% \text{ or } 1\%$
4. Test statistics

We know that

$$Z = \frac{P_1 - P_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$Q = 1 - p$$

$$p = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

$$p = \frac{1000 \times \frac{800}{1000} + 1200 \times \frac{800}{1200}}{1000 + 1200}$$

$$p = \frac{1600}{2200} = 0.7272$$

$$Q = 1 - 0.723 = 0.277$$

1) $H_0: P$ (There is no difference)

2) $H_1: P \neq P$ (There is difference)

3) $\alpha = 5\%$

4) Test statistics

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \quad (1 - P = 1 - 0.85 = 0.15)$$

$$= \frac{0.9 - 0.15}{\sqrt{\frac{(0.85)(0.15)}{20}}} = 0.630$$

$$\sqrt{\frac{(0.85)(0.15)}{20}}$$

5) Critical region Table value = 1.96

6) Conclusion

Calculated value < Table value

$$0.630 < 1.96$$

$\therefore H_0$ is accepted

2) A manufacturer user of light bulbs claims that an average of 2% of the bulbs manufactured by him are defective. A random sample of 400 bulbs contained 13 defective bulbs on the basis of the sample can you support manufacturer's claim at 5% loss?

Sol Given $p = \frac{13}{400}$, $P = \frac{2}{100}$

$$p = 0.0325, P = 0.02$$

$$\alpha = 5\%, n = 400, Q = 1 - P = 1 - 0.02 = 0.98$$

1) $H_0: p = P$

3) $\alpha = 5\%$

2) $H_1: p \neq P$

$$Z = \frac{\frac{800}{1000} - \frac{800}{1200}}{\sqrt{(0.423)(0.277)\left(\frac{1}{1000} + \frac{1}{1200}\right)}}$$

$$Z = \frac{0.8 - 0.66}{\sqrt{0.2003(0.001 + 0.00083)}}$$

$$Z = \frac{0.14}{\sqrt{0.2003(0.00183)}}$$

$$Z = \frac{0.14}{\sqrt{0.000367}}$$

$$Z = \frac{0.14}{0.12} = \frac{0.14}{0.0191}$$

$$Z = 1.1666$$

5. Critical region - table value = 2.58

Conclusion value < Table value

$$2.33 < 2.58$$

H_0 is accepted

Properties based on Single proposition

1. 20 people were attacked by a disease and only 18 survived will you reject the hypothesis that the survival rate if attacked by this disease is 85%. is there of the hypothesis that is more at 5% level?

Sol Given $p = \frac{18}{20}$, $P = 85 = \frac{85}{100} = 0.85$

$$\alpha = 5\% = 1.96, n = 20$$

4) Confidence Interval for μ , σ known

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right)$$

The confidence interval for 98%.

95%.

99%.

5) Confidence limit for population mean (μ) & population proportion & Estimation proportion mean.

→ 93% Confidence limit are $\bar{x} \pm 1.96$.

→ 99% Confidence limit are $\bar{x} \pm 2.58$.

→ 99.73% Confidence limit are $\bar{x} \pm 3$.

→ 98% Confidence limit are $\bar{x} \pm 2.33$.

→ 90% Confidence limit are $\bar{x} \pm 1.64$.

1. A random sample of size 81 was taken whose variance is 20.25 and mean is 32. Construct 98% Confidence interval.

Sol $n=81$

$$\text{Variance } (\sigma^2) = 20.25$$

$$\bar{x} = 32 \text{ (sample mean)}$$

$$\sigma = \sqrt{20.25} = 4.5$$

We know the 98% Confidence interval

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right)$$

$$\left(32 - z_{\alpha/2} \cdot \frac{4.5}{\sqrt{81}}, 32 + z_{\alpha/2} \cdot \frac{4.5}{\sqrt{81}} \right)$$

$$z_{\alpha/2} = 2.33 \text{ (98\%.)}$$

$$\left(32 - 2.33 \left(\frac{4.5}{\sqrt{81}} \right), 32 + 2.33 \left(\frac{4.5}{\sqrt{81}} \right) \right)$$

Parameter: The statistical measures such as mean and variance computed from sample observation referred as static ~~standard error~~

Standard Error: The standard deviation of sampling of a static is known its standard error

*Estimation

Estimation: The procedure (or) rate to determine an unknown population parameter is called as Estimation.

Types of Estimation:

→ PT Estimation

→ Interval Estimation

Formula:

→ maximum error of estimate

$$E_{\max} = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = Z_{\alpha/2} \cdot \bar{x} \quad \text{where } \bar{x} = \frac{\sigma}{\sqrt{n}}$$

→ Confidence $(1-\alpha) 100\%$.

→ To find n (Sample Size)

a) When α, E, σ are unknown the sample size is given by

$$n = \left(\frac{Z_{\alpha/2}}{E} \right)^2 (PQ)$$

$$n = \left(\frac{Z_{\alpha/2} \sigma}{E} \right)^2$$

b) When α, E, σ are known the sample size n is given by $E_{\max} = Z_{\alpha/2} \left(\sqrt{\frac{PQ}{n}} \right)$

4) Test statistics

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.0325 - 0.02}{\sqrt{\frac{0.02 \times 0.98}{400}}} = 1.7857$$

5) Critical region Table value = 1.96

6) Conclusion

cal Table value < table value

$$1.7857 < 1.96$$

$\therefore H_0$ is accepted

Definitions

1. Type 1: The hypothesis is true but our test is rejects it.

Type 2: The hypothesis is false but our test is accepted it.

2. Critical region: A region in the sample space with amounts to rejection of H_0 is termed as critical region or rejection.

3. Null hypothesis: For applying the test of significance we first set up of a hypothesis a definite statement about the population parameters such a hypothesis of no-difference and it is called null hypothesis (H_0).

Alternative hypothesis: Any hypothesis which is complementary to the null hypothesis is called an alternative hypothesis is denoted by H_1 .

5) What is the maximum error one can expect to make with probability 0.90, which using the mean of a random sample of size $n=64$ to estimate the mean of population with $\sigma^2 = 2.50$

Sol $n=64$

$$\text{Var} = \sigma^2 = 2.50$$

$$\text{S.D} = \sigma = \sqrt{2.50} = 1.6$$

$$\text{probability value} = 0.90 = 90\%$$

$$\text{maximum error} = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$z_{\alpha/2} = 1.645 \text{ (table value at 90\%)}$$

$$= (1.645) \left(\frac{1.6}{\sqrt{64}} \right)$$

$$= (1.645) \left(\frac{1.6}{8} \right)$$

$$= (1.645)(0.2) = \underline{0.329}$$

6) Assuming that $\sigma = 20.0$ how large a random sample data be taken the assert with probability 0.95 that the sample mean will not differ from the true mean by more than 3.0 units.

Sol Given $\sigma = 20.0$

$$\text{maximum error} = 3.0$$

$$\text{probability value} = 0.95 = 95\% = 1.96 = z_{\alpha/2}$$

Sol) Given

$$\mu = 11795 \quad S.D = 14054$$

$$\bar{x} = 11795 \quad \sigma = 14054$$

$$n = 50$$

$$95\% \text{ Confidence level} = z_{\alpha/2} = 1.96$$

We know that Confidence Interval formula

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$\left(11795 - \frac{(1.96)(14054)}{\sqrt{50}}, 11795 + \frac{(1.96)(14054)}{\sqrt{50}} \right)$$

$$\left(11795 - \frac{27545.84}{7.071}, 11795 + \frac{27545.84}{7.071} \right)$$

$$(11795 - 3895.60, 11795 + 3895.60)$$

$$(7899.4, 15690.6)$$

4) A random Sample of size 100 has a S.D of 5 what can you say about the maximum error with 95% confidence

Sol Given Sample of size $n = 100$

$$S.D = \sigma = 5$$

$$\text{WKT maximum error} = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$95\% = z_{\alpha/2} = 1.96$$

$$\text{maximum error} = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$= (1.96) \left(\frac{5}{\sqrt{100}} \right)$$

$$= (1.96) \left(\frac{5}{10} \right)$$

$$= \underline{0.98}$$

$$(32 - 2 \cdot 33(0.5), 32 + 2 \cdot 33(0.5))$$

$$(32 - 1.1625, 32 + 1.1625)$$

$$(\underline{30.835}, 33.165)$$

2) Find 95% Confidence limits for the mean of a normal distributed population from which the following sample was taken 15, 17, 10, 18, 16, 9, 7, 11, 13, 14

$$\bar{x} = \frac{15 + 17 + 10 + 18 + 16 + 9 + 7 + 11 + 13 + 14}{10}$$

$$\bar{x} = \frac{130}{10}, \bar{x} = 13$$

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{40}{9} = \sqrt{\frac{40}{9}} = 2.10$$

$$S = 2.10$$

95% of Confidence interval $z_{\alpha/2} = 1.96$

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \right)$$

$$\left(13 - 1.96 \cdot \frac{\sqrt{40}}{\sqrt{10}}, 13 + 1.96 \cdot \frac{\sqrt{40}}{\sqrt{10}} \right)$$

$$(15.26, 10.74)$$

3) The mean and standard deviation of a population are 11.795 and 14.054 respectively. what can one assert with 95% Confidence about the maximum error if $\bar{x} = 11.795$ & $n = 50$ and also construct 95% Confidence interval for the true mean

$$n=300, \sigma^2=225, \sigma=\sqrt{225}=15$$

$$\bar{x}=54, z_{\alpha/2}=1.96$$

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$\left(54 - \frac{1.96(15)}{\sqrt{300}}, 54 + \frac{1.96(15)}{\sqrt{300}} \right)$$

$$\left(54 - 1.96 \left(\frac{15}{17.320} \right), 54 + 1.96 \left(\frac{15}{17.320} \right) \right)$$

$$(54 - 1.96(0.8660), 54 + 1.96(0.8660))$$

$$(54 - 1.69736, 54 + 1.69736)$$

$$(\underline{52.3}, 55.697)$$

b) If Sample Size $n=144$ standard deviation $\sigma=4$ and the mean 150. Then 95% confidence interval is

$$\text{Sol)} n=144, \sigma=4, \bar{x}=150, z_{\alpha/2}=1.96$$

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$\left(150 - 1.96 \frac{4}{\sqrt{144}}, 150 + 1.96 \frac{4}{\sqrt{144}} \right)$$

$$\left(150 - 1.96 \frac{4}{12}, 150 + 1.96 \frac{4}{12} \right)$$

$$(150 - 1.96(0.33), 150 + 1.96(0.33))$$

$$(150 - 0.6468, 150 + 0.6468)$$

$$(\underline{149.35}, \underline{150.6468})$$

8) What is the size of the smallest sample required to estimate an unknown population to within a maximum error of 0.06 with at least 95% Confidence

Sol Given, maximum Error = $E = 0.06$

Given 95% Confidence Interval $= 1.96 = z_{\alpha/2}$
to find n

$$n = \text{maximum error} = E = z_{\alpha/2} \sqrt{\frac{pq}{n}}$$

Given unknown population

$$\text{Take } p = 1/2 \quad q = 1/2$$

Now

$$0.06 = (1.96) \sqrt{\frac{(0.5)(0.5)}{n}}$$

$$0.06 = (1.96) \sqrt{\frac{1/2 \times 1/2}{n}}$$

$$0.06 = (1.96) \sqrt{\frac{0.25}{n}}$$

$$0.06 = 1.96 \sqrt{\frac{1}{4n}}$$

Squaring both sides

S.O.Bs

$$(0.06)^2 = (1.96)^2 \frac{0.25}{n}$$

$$0.36 = 3.8416 \left(\frac{1}{4n} \right)$$

$$(0.036) = (3.8416) \frac{0.25}{n}$$

$$n = \frac{3.8416}{4 \times 0.36}$$

$$n = \frac{(3.8416)(0.25)}{0.036}$$

$$n = 266.77$$

$$n \approx \underline{\underline{267}}$$

$$n = 266.77 \approx 267$$

9) A sample of size 300 was taken whose Variance is 225 & mean 54 Construct 95% Confidence Interval for the mean

to find n value:

$$\begin{aligned}\text{We know that } n &= \left(\frac{Z_{\alpha/2} \cdot \sigma}{E} \right)^2 \\ &= \left(\frac{1.96 \times \frac{20.0}{3}}{0.05} \right)^2 \\ &= 170.74 \approx 171\end{aligned}$$

4) If we can assert with 95%. that the maximum error is 0.05 and $p=0.2$ find the size of the sample to find n

Sol: max error = 0.05

probability value $p=0.2$

$$q = 1 - p = 1 - 0.2 = 0.8$$

$$95\% \text{ of confidence level} = Z_{\alpha/2} = 1.96$$

We know that maximum error = $Z_{\alpha/2} \sqrt{\frac{pq}{n}}$

$$0.05 = (1.96) \left(\sqrt{\frac{(0.2)(0.8)}{n}} \right)$$

$$0.05 = (1.96) \sqrt{\frac{0.16}{n}}$$

Squaring on both sides

$$(0.05)^2 = (1.96)^2 \left(\frac{0.16}{n} \right)$$

$$0.0025 = \frac{(3.8416)(0.16)}{n}$$

$$n = \frac{(3.8416)(0.16)}{0.0025}$$

$$n = \frac{0.614}{0.0025}$$

$$n = 245.86 \approx 246$$